03 Using inverse rate plots to size reactors

Why are we learning this?

In the previous lesson, we rewrote the reactor mole balances in terms of conversion. Today, we will use those conversion-based balances to visualize reactor design graphically using **Levenspiel (Inverse Rate) Plots**.

Rather than solving equations immediately, Levenspiel plots allow us to:

- estimate reactor volume,
- compare reactor types,
- understand how conversion affects reactor sizing,
- evaluate reactor sequences graphically.

By the end of today's lesson, you should understand:

1. What a Levenspiel Plot represents.
2. Why $\frac{F_{A0}}{-r_A}$ appears on the vertical axis.
3. How to determine PFR volume graphically.
4. How to determine CSTR volume graphically.
5. Why reactor type affects the volume required to reach a desired conversion.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)

↓
►Graphical Reactor Design◄
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Today's Location: **Levenspiel Plots**

Today we connect reactor design equations to graphical reactor design.

A Levenspiel Plot or Inverse Rate Plot plots the quantity $F_{A0}/-r_A$ versus X_A. These plots are used for visualizing the CSTR and PFR volumes needed to achieve a certain conversion for a reaction. They are also used for reactor sequencing.

Example of a Levenspiel Plot:

If $-r_A = kC_A^n$, where $n > 0$,

In other words, the rate gets

Hence, $\frac{1}{kC_A^n}$ gets

What Happens as Conversion Increases?

For many reactions:

- reactant concentration decreases with conversion,
- reaction rate decreases,
- achieving additional conversion becomes harder.

As the rate becomes smaller, $\frac{F_{A0}}{-r_A}$ becomes larger. This is why many Levenspiel plots rise with increasing conversion.

Engineering Interpretation: The last 10% conversion often requires much more reactor volume than the first 10%.

How do we get the volume of a PFR using a Levenspiel Plot?

PFR mole balance: $\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$, hence $V =$

where r_A is a function of

Hence, the PFR volume is

Why Does Area Matter?

The PFR design equation is

$$\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$$

Integrating means: Add up many tiny pieces of reactor volume. Graphically, adding up those tiny pieces corresponds to finding the area under the curve.

This is why:

PFR volume = area under the Levenspiel curve.

Interpreting PFR Design

Every small increase in conversion contributes a small amount of reactor volume. The total reactor volume is simply the accumulation of all those contributions. Visually:

Area Under Curve = PFR Volume

How do we get the volume of a CSTR using a Levenspiel Plot?

CSTR mole balance: $V = \frac{F_{A0}X_A}{-r_A}$, hence $V =$

Hence, the CSTR volume is

Why Is a CSTR a Rectangle?

A CSTR is perfectly mixed. As a result:

Conditions throughout the reactor equal outlet conditions.

The reactor therefore operates at a single reaction rate. Since the rate is constant throughout the reactor volume, the graphical representation becomes a rectangle.

Interpreting the CSTR Rectangle

Rectangle Width = ΔX. Represents the conversion achieved.

Rectangle Height = $\frac{F_{A0}}{-r_A}$. Represents the reaction conditions at the outlet.

Rectangle Area, V = represents reactor volume.

Big Ideas

The reactor design equations derived previously can be interpreted as geometric areas on a graph. For reactor design:

- PFR volume = area under a curve
- CSTR volume = area of a rectangle

Because areas are easy to visualize, Levenspiel plots provide insight that can be difficult to obtain from equations alone.

What Does a Levenspiel Plot Show?

A Levenspiel plot links:

- reaction kinetics,
- conversion,
- reactor volume

on a single graph.

The horizontal axis tells us:

How far the reaction has progressed.

The vertical axis tells us:

How much reactor is needed to achieve additional conversion.

Large values of $\frac{F_{A0}}{-r_A}$ indicate that conversion is becoming more difficult.

Why Use a Levenspiel Plot?

Levenspiel plots provide quick insight about:

- How large should the reactor be?
- Which reactor type should I use?
- Should reactors be placed in series?
- How much additional conversion is worth pursuing?

If you have a PFR and a CSTR which are both the same Volume, which one will give the higher conversion for this reaction?

Why?

A question to ponder: Why might $F_{A0}/-r_A$ decrease with X_A? We will come back to this when we discuss rate laws and the rate constant.

Summary

Today we used conversion-based design equations to create Levenspiel plots.

We learned that:

- A Levenspiel plot graphs $\frac{F_{A0}}{-r_A}$ versus conversion.
- PFR volume is represented by the area under the curve.
- CSTR volume is represented by the area of a rectangle.
- Graphical design provides intuition about reactor performance.
- Reactor type affects the volume required to achieve a desired conversion.
- For reactions where the rate decreases as conversion increases, PFRs use their volume more efficiently and can achieve higher conversion than an equal volume CSTR.

Key Takeaways

☑ **A Levenspiel Plot combines kinetics and reactor design.** The plot shows how reaction rate changes affect reactor size.

☑ **PFR volume is an area under a curve.** $V_{PFR} = \int \frac{F_{A0}}{-r_A} dX$

☑ **CSTR volume is a rectangle.** $V_{CSTR} = \frac{F_{A0}X}{-r_A}$

☑ **Rising curves indicate that conversion is becoming more difficult.** Higher conversion often requires disproportionately larger reactors.

☑ **Most important idea.** A Levenspiel plot transforms reactor design equations into geometry. Reactor volumes become areas, allowing us to visualize reactor behavior, compare reactor types, and gain quick insight about reactor sequences.